

Campus Based Donation Management System

B.Sc. Engineering Final Defense Presentation

Ramjan Ali Kha

Student ID: 200114

Registration Year: 2020-21

Department of Computer Science and Engineering
Jashore University of Science and Technology
Jashore-7408, Bangladesh

Supervisor:

Dr. Mohammad Nowsin Amin Sheikh
Assistant Professor

April 27, 2026

Presentation Outline

Presentation Outline

Introduction

- **Background:** Modern universities generate significant amounts of resources that can be repurposed, but lack an organized system to facilitate donations.
- **What is the DMS?** The Campus Based Donation Management System (DMS) is a centralized, digital platform connecting donors (students, faculty) with recipients in need.
- **Why it matters:** Promotes a culture of sustainability, resource-sharing, and community support within the campus ecosystem.

Presentation Outline

Problem Statement

Current Challenges in Campus Resource Sharing

- **Lack of Coordination:** No centralized platform for students/faculty to donate usable goods (books, electronics, clothing).
- **Inefficiency:** Information relies on fragmented social media groups or word of mouth.
- **Transparency Issues:** Donors often do not know if their items reached those genuinely in need.
- **Tracking:** Administrators struggle to track campaigns and overall impact.

Presentation Outline

Objectives

The primary objectives of the Campus Based Donation Management System are:

- ➊ **Centralized Platform:** Develop a robust web application for managing campus-wide donations seamlessly.
- ➋ **Campaign Management:** Allow administrators to run targeted donation campaigns (e.g., winter clothing drives, book banks).
- ➌ **Transparency & Tracking:** Provide real-time tracking of donations from pledge to delivery.
- ➍ **User-Friendly Experience:** Create a highly responsive, accessible, and intuitive UI for both donors and admins.
- ➎ **Analytics:** Deliver insight via a dashboard to measure community impact and engagement.

Presentation Outline

System Architecture & Tech Stack

Frontend (Client-Side)

- React with TypeScript
- Redux Toolkit for State Management
- Tailwind CSS for UI Design

Backend (Server-Side)

- .NET Core Web API
- Entity Framework Core
- RESTful API Architecture

Architectural Highlights

- Role-Based Access Control (Admin vs. Donor)
- JWT-based Authentication
- Responsive Design implementation for mobile and desktop
- Separation of concerns (Frontend / API / Database)

Presentation Outline

Key Features of the System

- Donor Management** Secure registration, profile management, and donation history tracking for users.
- Item Posting & Claiming** Donors can list items; verified users can request/claim them.
- Campaign Dashboard** Dedicated admin interface to monitor active campaigns, progress bars, and total impact.
- Analytics & Reporting** Visual dashboards illustrating donation trends across the university.

Presentation Outline

Implementation Snapshot

(Include screenshots of the application here during the final defense)

- **Landing Page:** Showcasing active campaigns and recent donations.
- **Donor Dashboard:** Illustrating user history and impact score.
- **Admin Panel:** Demonstrating campaign creation and user role management.

Presentation Outline

Conclusion & Future Work

Conclusion:

- Successfully built and deployed a scalable web-based solution for campus resource management.
- Bridges the gap between surplus resources and student needs efficiently and transparently.

Future Work:

- **Mobile Application:** Develop a native mobile app companion for push notifications.
- **AI Integrations:** Suggest items to students based on predictive needs.
- **Inter-University Network:** Expand the scope to include collaborations with neighboring universities.

Thank You!

Any Questions?